SMART INDIA HACKATHON 2026

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- **Problem Statement ID -** SIH26261
- **Problem Statement Title -** AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- **Theme -** Smart Automation
- **PS Category -** Software
- **Team ID -** 140444
- **Team Name -** Geo Pulse Shifters



Geo Pulse Shifters

SMART VERIFY

(Smart Scrutiny & Regulatory Verification for Technical Education)

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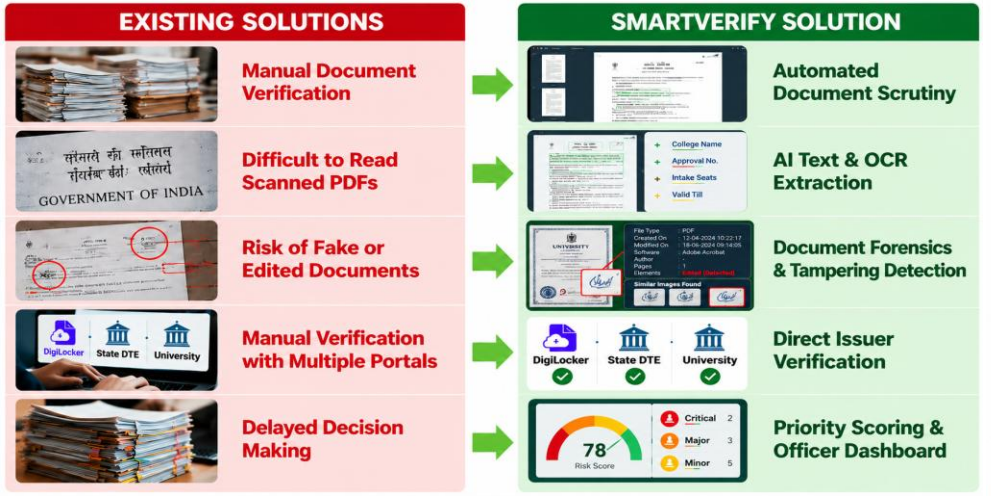
IDEA / SOLUTION

SMARTVERIFY is an **AI-powered scrutiny assistant combining AICTE rule checks** with **document forensics** to help officers validate college applications in **real time**.

Advanced Features:

- **Automated Document** Scrutiny with **evidence-linked findings** on an interactive officer dashboard.
- **Grounded VLM & Agentic RAG (Qwen2-VL + LangGraph)** for digital and scanned PDFs.
- **Tampering & Forgery Detection** catching edited certificates, altered metadata, and reused photos.
- **Smart Knowledge Graph linking** college claims (intake seats, faculty, labs) to official baselines.
- Direct Issuer **Verification validating records** with DigiLocker, State DTE, and Universities via QR/signatures.
- **Explainable Priority Scoring** categorizing issues by **severity (critical, major, minor)** for fast review.
- **Offline-Ready & Secure Design** using zero paid APIs, tamper-evident audit logs, and human-in-the-loop control.

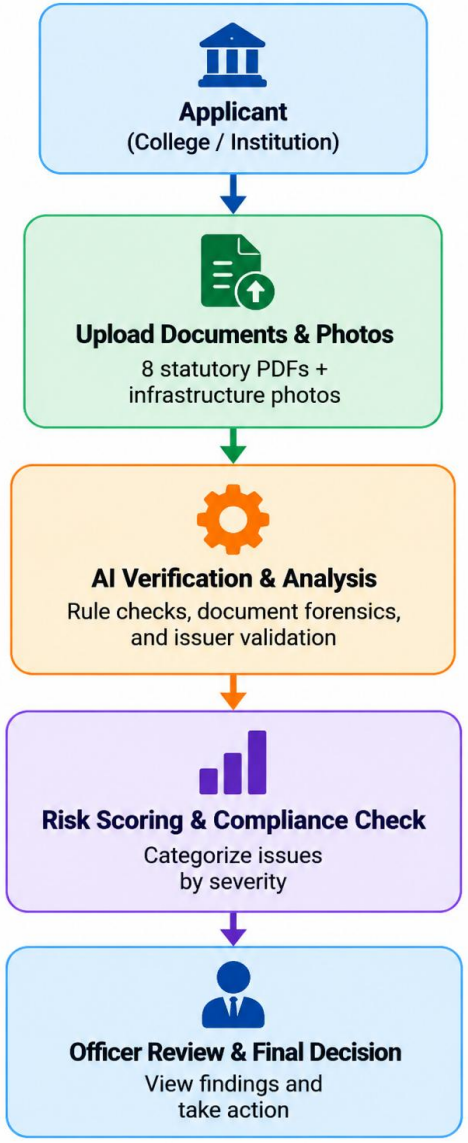
HOW SMART VERIFY ADDRESSES THE PROBLEM



UNIQUE VALUE PROPOSITION

- **Direct Source Verification** — Auto-checks certificates against **DigiLocker, State DTE, Universities** via QR, signatures.
- **Tamper-Proof Audit Trail** — Logs every check and action in an unalterable ledger, ensuring audit transparency.
- **Instant Deficiency Alerts** — Catches missing, expired, invalid documents upon upload, helping colleges fix errors early.
- **Automated Infrastructure Triage** — Inspects campus, lab photos with computer vision, filtering blurry, fake proofs.

USER FLOW



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TECHNICAL APPROACH

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ARCHITECTURE FLOW

INGESTION & FORENSIC ALTERATION SCREENING

Application Input

PDF

Institution Dossier
(Certificates, Approvals, etc.)

Python | PyMuPDF — PDF Alteration Check (PyMuPDF)

Alteration? — No → continue

Yes → Flag for Human Scrutiny & Physical EVC Inspection

Agentic RAG + VLM : Autonomous Statutory Verification Pipeline

Orchestrator Agent (LangGraph / CrewAI)

✓ Plan ✓ Execute ✓ Verify ✓ Self-Correct

VLM Spatial Grounding & Document Classifier

Qwen2-VL | Florence-2

FIRE SAFETY CERTIFICATE
Valid Till : 2027-06-30

- Classifies document type (e.g., Fire Safety NOC)
- Detects seals, signatures, and key fields
- Returns text + bounding box coordinates

Agentic RAG Regulatory Engine

LangGraph | ChromaDB | LlamaIndex

AICTE APH 2024–27 (= 150 Pages)

- Multi-hop retrieval across AICTE APH 2024–27
- Generates dynamic statutory checklist (clause-wise)
- Cross-verifies extracted details (e.g., validity, area, faculty, seats)

Deterministic Reconciler & Decision Dashboard

FastAPI | PostgreSQL

- Mathematical date & quota checks (e.g., intake, land area)
- Claim-to-Evidence Knowledge Graph (document ↔ clause ↔ evidence)
- Decision output with Human-in-the-Loop review

Self-Correction Loop | If any field is missing or unclear, the agent replans (e.g., "Re-inspect page 2 at higher resolution").

Statutory Scrutiny Outcomes & Governance

1 Prioritization & Risk Gauge (Score: 0 – 100)

- Low (0 – 30) : Fast-Track Queue
- Medium (31 – 70) : Routine Scrutiny
- High (71 – 100) : Escalated Inspection Priority

2 The 4 Official AICTE Decision-Support States

- Ready for Scrutiny (100% Deterministic Pass)
- Awaiting Rectification (Notice Sent to College)
- Physical EVC Inspection Recommended
- Under Scrutiny (Officer In-Progress)

3 Legal Traceability & Audit Export

- SHA-256 Hash-Chained Audit Ledger (Tamper-Proof)
- Instant Appellate Dossier Export for Court / RTI

4 Instant Multi-Channel Alerts

- Automated SMS / Email to College & Regional Officer

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FEASIBILITY AND VIABILITY

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FEASIBILITY

Combines **PyMuPDF with Lightweight VLM (Florence-2) & ChromaDB Vector Store.**

Runs **AICTE APH rules in YAML over PostgreSQL**, ensuring 100% explainable checks with zero AI hallucinations.

Uses **OpenCV and pHash** to audit facility photos and catch duplicate certificates without requiring GPUs.

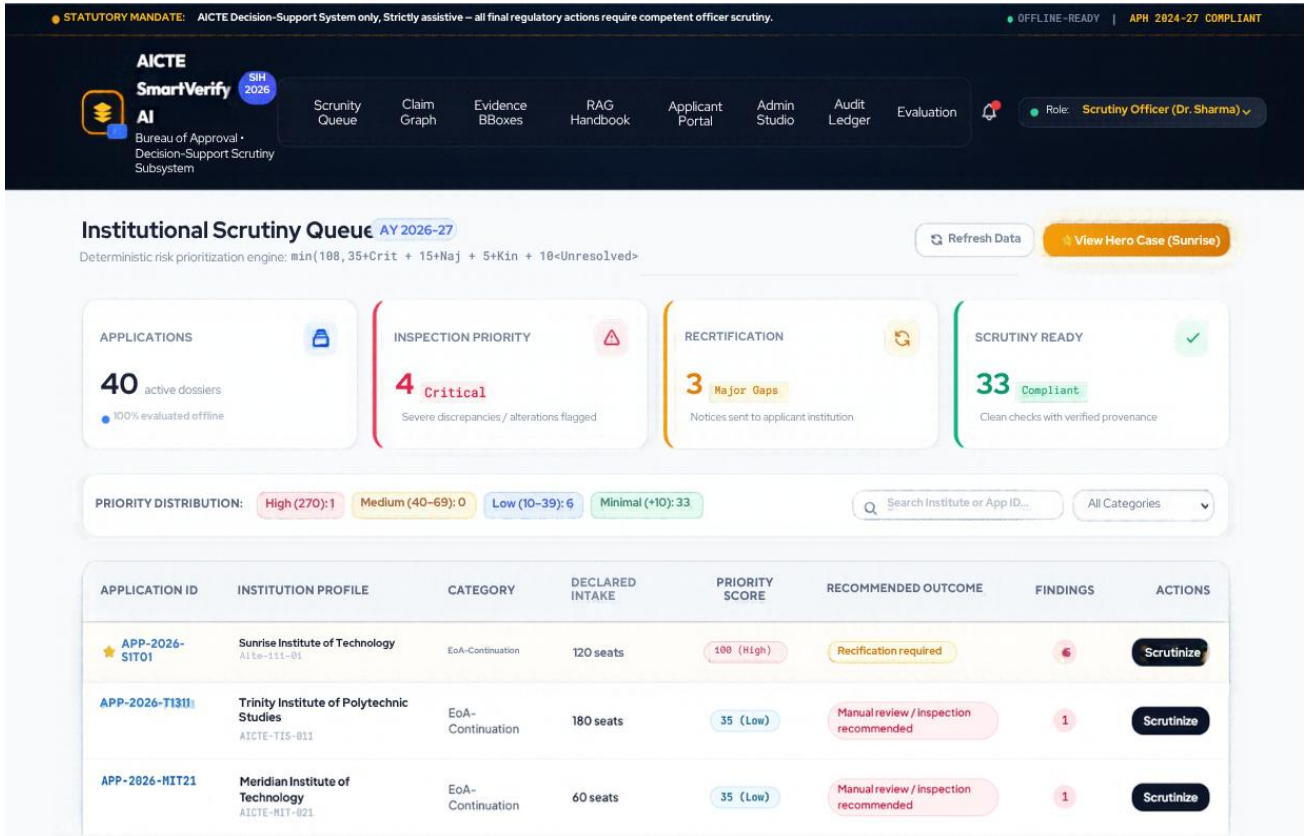
VIABILITY

Integrates directly with the **AICTE portal**, leaving all final approvals strictly with human officers.

Automatically **masks Aadhaar, PAN, and phone numbers**, ensuring strict data privacy compliance.

Built on **free open-source tools with zero paid APIs**, eliminating recurring licensing costs for AICTE.

DEMO UI



POTENTIAL CHALLENGES

Poor quality scans and mobile photos from rural colleges can reduce OCR accuracy and trigger false flags.

Issuing authority portals (State DTE, Universities) may experience network downtime during peak submission weeks.

OVERCOME METHODS

OpenCV preprocessing (deskew/denoise) with VLM spatial grounding and agentic self-correction for degraded scans.

Asynchronous caching and local QR/digital signature verification, ensuring uninterrupted scrutiny even during network outages.

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IMPACTS AND BENEFITS

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TECHNICAL INSTITUTIONS & APPLICANTS

Detects **missing, expired, or inconsistent documents at the upload stage**, giving colleges **real-time rectification guidance** before official scrutiny instead of facing sudden rejections, delayed approvals, or repeated manual submissions.

AICTE OFFICERS & SCRUTINY COMMITTEES

Provides an **interactive evidence-linked decision dossier** with word-level bounding boxes and explainable risk scores, shifting scrutiny from **slow, repetitive page-by-page manual verification** to **fast, high-confidence statutory oversight**.

NATIONAL EDUCATION ECOSYSTEM

Built on **offline-ready open architecture** aligned with the AICTE APH 2024–27, SMARTVERIFY is designed for **zero-cost, secure deployment across all 10,000+ technical institutions in India**, eliminating forged affiliations and ghost infrastructure nationwide.

Fair, evidence-linked scrutiny without arbitrary rejections

Complete protection against recycled certificates and ghost faculty

Potential Impact on the Target Audience:

1. FASTER APPLICATION SCRUTINY

Dossier review time cut from **45 minutes to under 5 minutes**.

2. ELIMINATED FORGERY & TAMPERING

Digital forensics catches **altered dates, cloned NOCs, and edited photos**.

3. STANDARDIZED REGULATORY DECISIONS

Deterministic rule engine ensures **consistent, unbiased statutory evaluations**.

4. AUDITABLE ACCOUNTABILITY

Immutable SHA-256 ledger provides **transparent proof for CAG audits and appeals**.

Benefits of the Solution:

Scrutiny Efficiency
82% faster review time

Explainable AI
Evidence-grounded scoring

Zero Forgery Leakage
ELA + pHash + Metadata checks

National Scalability
10,000+ technical colleges



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RESEARCH AND REFERENCES

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AICTE Approval Process Handbook (2024–27). Statutory norms for intake, land, and faculty quotas. *Ministry of Education, GoI.* — Click here

Digital Document Forensics (Krawetz, IEEE). Error Level Analysis (ELA) and perceptual hashing (pHash) for image tampering detection. — Click here

Regulatory Knowledge Graphs (Agal et al., 2025). Entity-relation graph mapping applied to AICTE claims vs. verified baselines. — Click here

DigiLocker

National standard for paperless verification. **SMARTVERIFY** validates NOCs directly via **Ed25519 digital signatures and QR codes**.

Click Here

AICTE APH Regulations

Statutory regulations for intake and faculty ratios. Encodes clauses into **deterministic YAML rules** for explainable, error-free checks.

Click Here

DPDP Act 2023

India's data privacy standard. Automatically **masks Aadhaar, PAN, and contact details** locally with zero cloud leaks

Click Here

GitHub Repo Click Here

Project Documents Click Here

ASPECTS	OTHERS (Manual / Traditional)	SMARTVERIFY
Scrutiny Speed	**45+ mins** — slow page-by-page checking	**Under 5 mins** automated extraction & validation
Forgery Check	**Zero forensic check** — misses edited dates/PDFs	**Multi-modal Forensics** — ELA, pHash & metadata audit
Cross-College Audit	**Isolated** — cannot spot reused certificates	**National Graph** — flags duplicates across colleges
Rule Consistency	**Subjective** — depends on individual officer memory	**Deterministic Rules** — 100% reproducible YAML checks
Data Privacy & Cost	**Paid cloud APIs** with privacy leak risk	**Zero-Cost & Offline** — local CPU, DPDP Act compliant
Decision Authority	**Binary black-box** auto-decisions	**Non-Binary Advisory** — strictly human-in-the-loop